PVsyst V8.0.15

PVsyst - Simulation report

Grid-Connected System

Project: Sia Project

Variant: Jinko 645W_0.5m_15tilt

Ground system (tables) on a terrain

System power: 3199 kWp

Siá - Cyprus

Author

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Project summary

Geographical Site	Situation		Project settings	
Siá	Latitude	34.94 °(N)	Albedo	0.20
Cyprus	Longitude	33.39 °(E)		
	Altitude	322 m		
	Time zone	UTC+2		

Weather data
Siá
Meteonorm 8.2 (2006-2013), Sat=100% - Synthetic

System summary

Grid-Connected System — **Ground system (tables) on a terrain**

Orientation #1		Orientation #2		Near Shadings	
Fixed plane		**Fixed plane**		According to strings : Fast (table)	
Tilt/Azimuth	15 / -90 °	Tilt/Azimuth	15 / 90 °	Electrical effect	100 %

System information

PV Array		Inverters	
Nb. of modules	4960 units	Nb. of units	18 units
Pnom total	3199 kWp	Total power	2700 kWac
		Pnom ratio	1.18

User's needs
Unlimited load (grid)

Results summary

Produced Energy	5049.6 MWh/year	Specific production	1578 kWh/kWp/year	Perf. Ratio PR	89.69 %
				Bifacial perf. ratio	81.37 %

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General parameters

Grid-Connected System

Ground system (tables) on a terrain

Orientation #1

Fixed plane		**Sheds configuration**		**Sizes**	
Tilt/Azimuth	15 / -90 °	Nb. of sheds	626 units	Sheds spacing	9.73 m
		Set of tables		Sensitive width	4.77 m
		Shading limit angle		GCR Shading	49.0 %
		Limit profile angle	14.3 °		

Orientation #2

Fixed plane		**Sheds configuration**		**Sizes**	
Tilt/Azimuth	15 / 90 °	Nb. of sheds	626 units	Sheds spacing	9.73 m
		Set of tables		Sensitive width	4.77 m
		Shading limit angle		GCR Shading	49.0 %
		Limit profile angle	14.3 °		

Models used		**Horizon**		**Near Shadings**	
Transposition	Perez	Average Height	3.8 °	According to strings : Fast (table)	
Diffuse	Perez, Meteonorm			Electrical effect	100 %
Circumsolar	separate				

Bifacial system definition

Orientation #1		**Orientation #2**	
Bifacial system		**Bifacial system**	
Model	Unlimited Sheds 2D Model	Model	Unlimited Sheds 2D Model
Bifacial model geometry		**Bifacial model geometry**	
Sheds spacing	9.73 m	Sheds spacing	9.73 m
Sheds width	4.77 m	Sheds width	4.77 m
Limit profile angle	14.3 °	Limit profile angle	14.3 °
GCR Bifacial	49.0 %	GCR Bifacial	49.0 %
Height above ground	1.50 m	Height above ground	1.50 m
Nb. of sheds	33 units	Nb. of sheds	33 units
Bifacial model definitions		**Bifacial model definitions**	
Ground albedo	0.30	Ground albedo	0.30
Bifaciality factor	80 %	Bifaciality factor	80 %
Rear shading factor	5.0 %	Rear shading factor	5.0 %
Rear mismatch loss	10.0 %	Rear mismatch loss	10.0 %
Shed transparent fraction	0.0 %	Shed transparent fraction	0.0 %

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module		**Inverter**	
Manufacturer	Jinkosolar	Manufacturer	Solis
Model	JKM-645N-66HL4-BDV	Model	S6-GC3P150K07-NV-ND
(Custom parameters definition)		(Original PVsyst database)	
Jinko_JKM_645N_66HL4_BDV.PAN		Unit Nom. Power	150 kWac
Unit Nom. Power	645 Wp	Number of inverters	126 * MPPT 14% 18 units
Number of PV modules	4960 units	Total power	2700 kWac
Nominal (STC)	3199 kWp	Operating voltage	160-1000 V
Modules	248 string x 20 In series	Pnom ratio (DC:AC)	1.18
At operating cond. (50°C)		No power sharing between MPPTs	
Pmpp	2968 kWp		
U mpp	767 V		
I mpp	3871 A		

PV Array Characteristics

Total PV power		**Total inverter power**	
Nominal (STC)	3199 kWp	Total power	2700 kWac
Total	4960 modules	Number of inverters	18 units
Module area	13398 m²	Pnom ratio	1.18
Cell area	12512 m²		

Array losses

Array Soiling Losses
Loss Fraction 2.0 %

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 27.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 2.2 mΩ
Loss Fraction 1.00 % at STC

LID - Light Induced Degradation
Loss Fraction 1.0 %

Module Quality Loss
Loss Fraction -0.75 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

Strings Mismatch loss
Loss Fraction 0.10 %

IAM loss factor
Incidence effect (IAM): User defined profile

0°	20°	40°	60°	65°	70°	75°	80°	85°
1.000	1.000	1.000	1.000	0.998	0.995	0.981	0.930	0.736

AC wiring losses

Inv. output line up to MV transfo

Inverter voltage	400 Vac tri
Loss Fraction	1.85 % at STC
Inverter: S6-GC3P150K07-NV-ND	
Wire section (18 Inv.)	Copper 18 x 3 x 120 mm²
Average wires length	108 m

MV line up to Injection

MV Voltage	20 kV
Wires	Copper 3 x 95 mm²
Length	1005 m
Loss Fraction	0.16 % at STC

AC losses in transformers

MV transfo

Medium voltage	20 kV
Transformer parameters	
Nominal power at STC	3.15 MVA
Iron Loss (24/24 Connexion)	2.78 kVA
Iron loss fraction	0.09 % at STC
Copper loss	35.80 kVA
Copper loss fraction	1.14 % at STC
Coils equivalent resistance	3 x 0.58 mΩ

Horizon definition

Horizon from Meteonorm web service, Lat=34°56'40", Long=33°23'25"

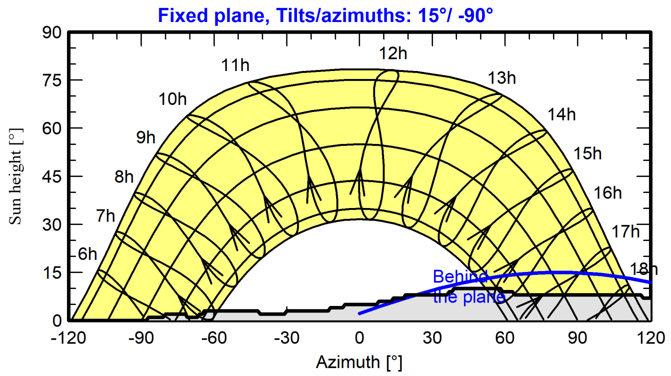
Average Height	3.8 °	Albedo Factor	0.73
Diffuse Factor	0.97	Albedo Fraction	100 %

Horizon profile

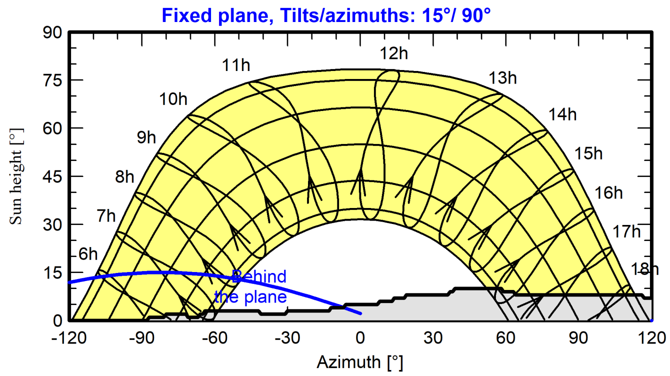
Azimuth [°]	-180	-172	-171	-168	-162	-161	-158	-157	-88	-87	-81	-80	-72	-71
Height [°]	0.0	0.0	1.0	1.0	2.0	1.0	1.0	0.0	0.0	1.0	1.0	2.0	2.0	1.0
Azimuth [°]	-67	-66	-65	-64	-42	-41	-31	-30	-13	-12	-7	-6	7	8
Height [°]	1.0	2.0	2.0	3.0	3.0	2.0	2.0	3.0	3.0	4.0	4.0	5.0	5.0	6.0
Azimuth [°]	13	14	18	19	36	37	38	39	58	59	63	64	116	117
Height [°]	6.0	7.0	7.0	8.0	8.0	9.0	9.0	10.0	10.0	9.0	9.0	8.0	8.0	7.0
Azimuth [°]	120	121	123	124	129	130	133	134	146	151	152	173	174	179
Height [°]	7.0	6.0	6.0	5.0	5.0	4.0	4.0	3.0	3.0	2.0	1.0	1.0	0.0	0.0

Sun Paths (Height / Azimuth diagram)

Orientation #1

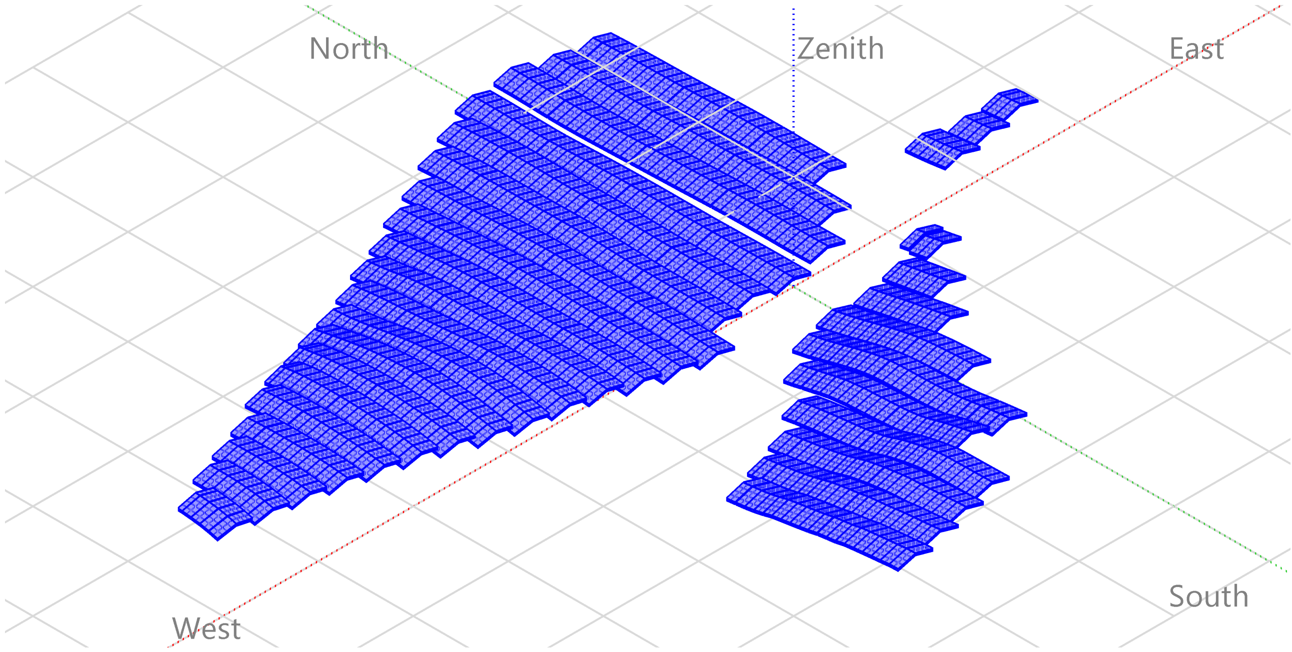


Orientation #2



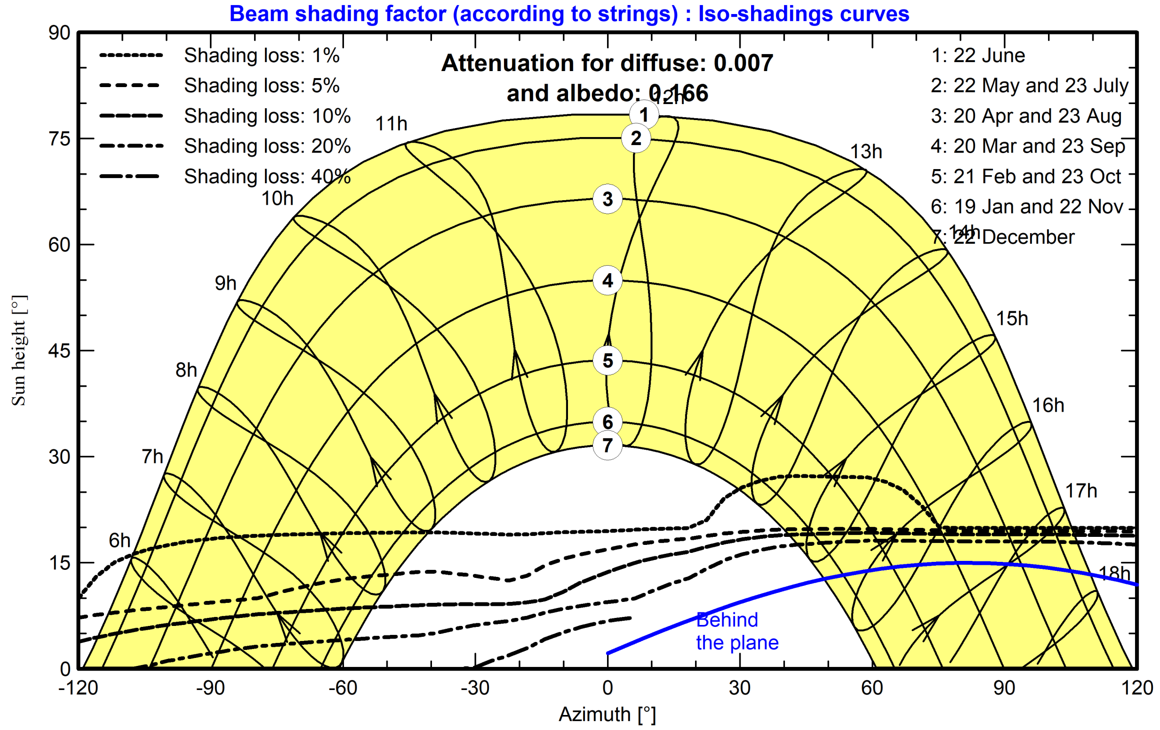
Near shadings parameter

Perspective of the PV-field and surrounding shading scene

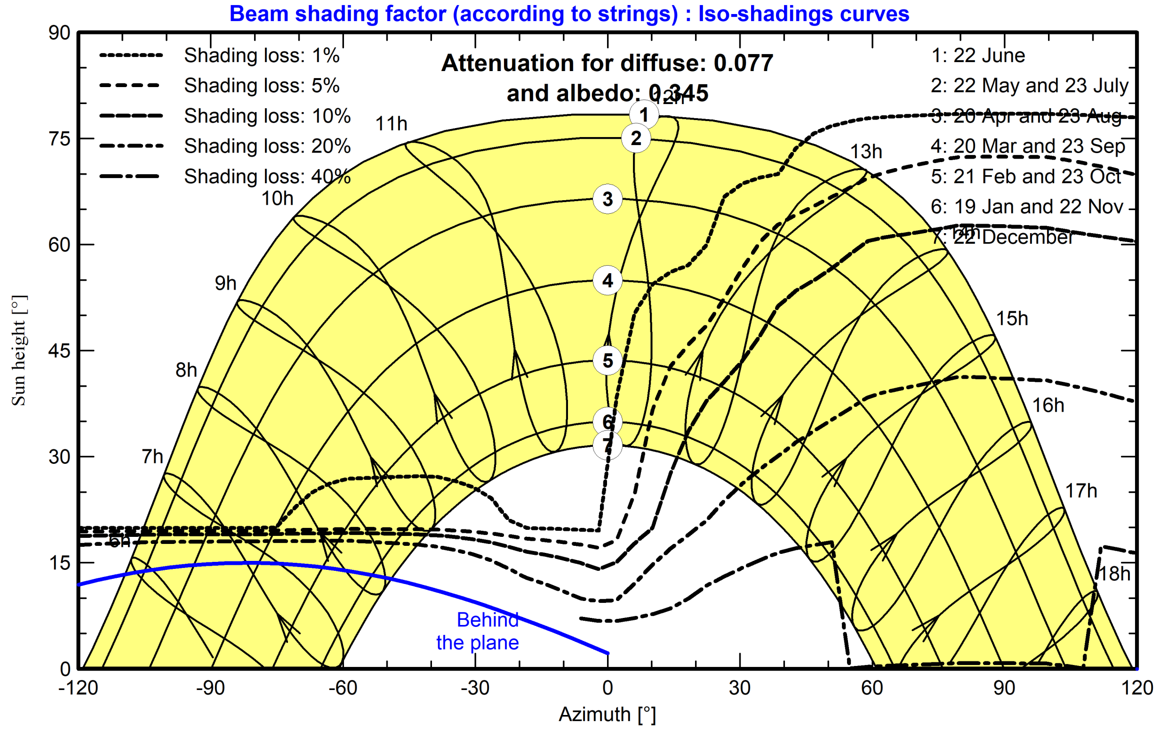


Iso-shadings diagram

Orientation #1 - Fixed plane, Tilts/azimuths: 15°/ -90°



Orientation #2 - Fixed plane, Tilts/azimuths: 15°/ 90°

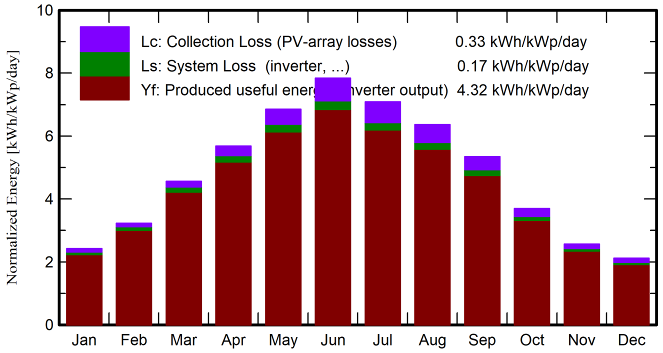


Main results

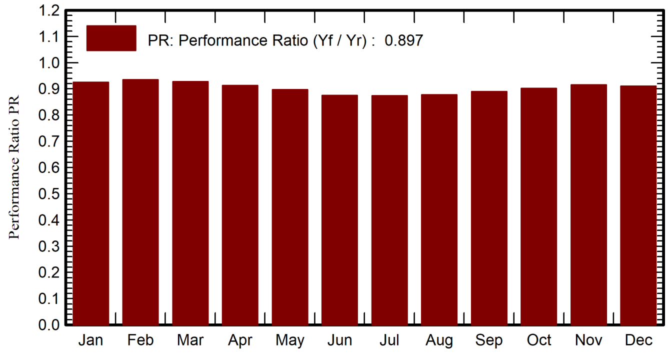
System Production

Produced Energy	5049.6 MWh/year	Specific production	1578 kWh/kWp/year
		Perf. Ratio PR	89.69 %
		Bifacial perf. ratio	81.37 %

Normalized productions (per installed kWp)



Performance Ratio PR



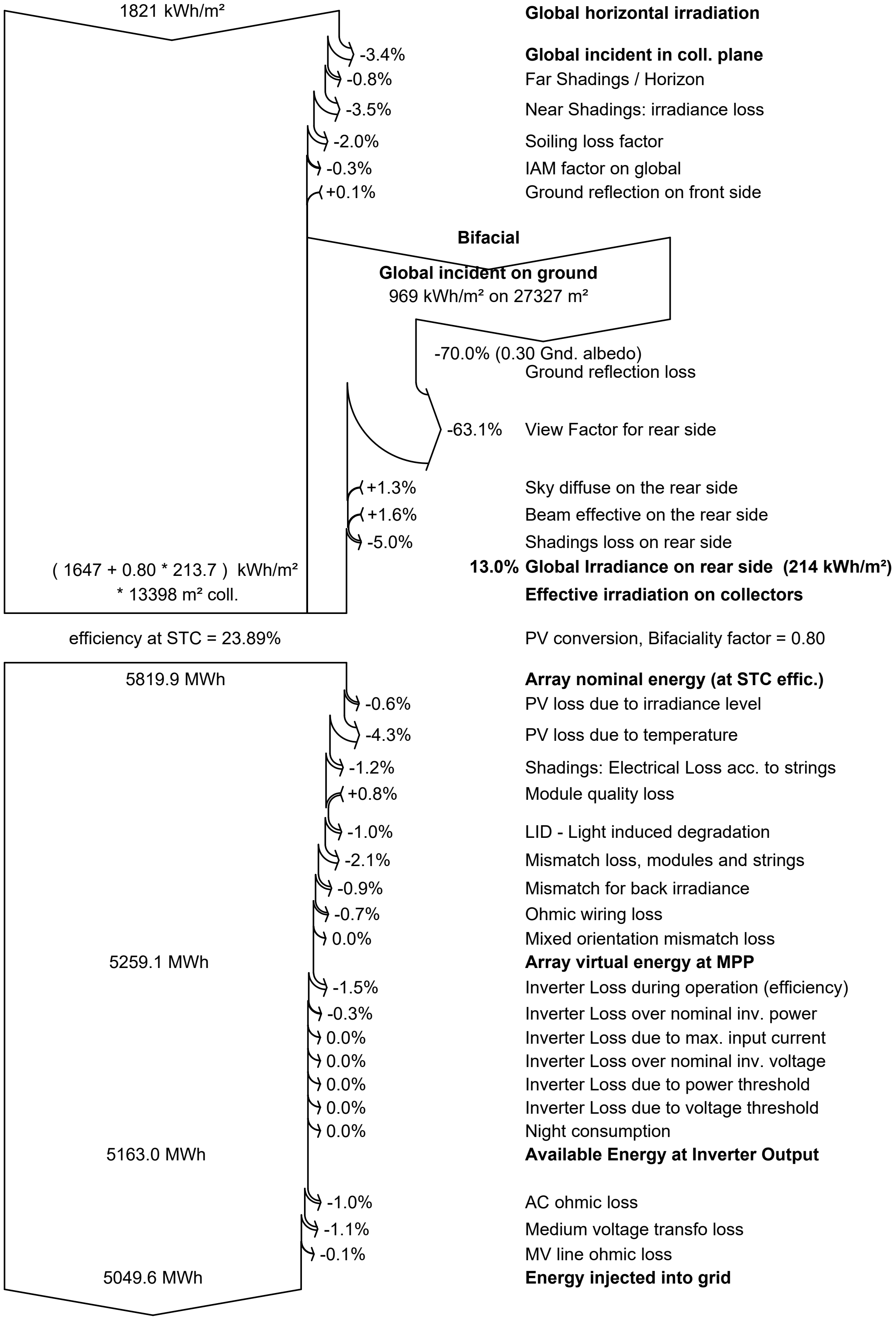
Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray MWh	E_Grid MWh	PR ratio	PRBifi ratio
January	79.7	34.95	10.26	75.1	68.4	230.8	222.3	0.926	0.831
February	94.8	39.74	10.21	90.3	83.6	280.1	270.0	0.935	0.843
March	147.4	58.09	13.02	141.3	132.3	435.3	419.2	0.927	0.840
April	175.7	66.33	16.72	170.6	159.9	517.4	497.9	0.912	0.828
May	217.2	76.58	21.98	212.4	200.5	633.2	609.3	0.897	0.817
June	240.5	61.01	26.22	235.2	223.0	684.4	658.0	0.874	0.797
July	224.7	74.62	29.72	219.7	207.7	638.4	614.5	0.874	0.796
August	203.1	70.77	29.44	197.4	185.7	575.6	554.2	0.877	0.798
September	166.5	55.12	25.70	160.5	150.2	474.2	456.9	0.890	0.806
October	119.9	51.46	22.19	114.6	106.1	342.5	330.4	0.901	0.814
November	82.0	37.04	16.11	77.1	70.9	234.1	225.7	0.915	0.824
December	69.9	30.26	11.83	65.6	59.2	198.5	191.1	0.911	0.817
Year	1821.3	655.98	19.51	1759.9	1647.4	5244.5	5049.6	0.897	0.814

Legends

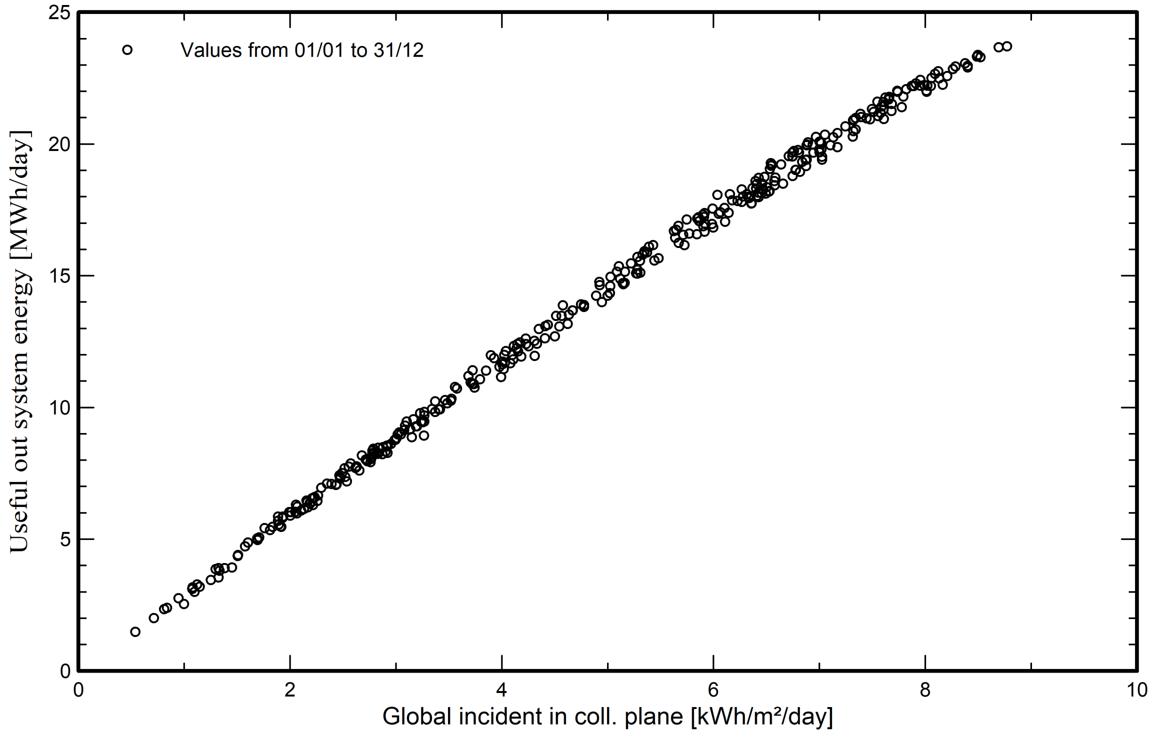
GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane	PRBifi	Bifacial Performance Ratio
GlobEff	Effective Global, corr. for IAM and shadings		

Loss diagram

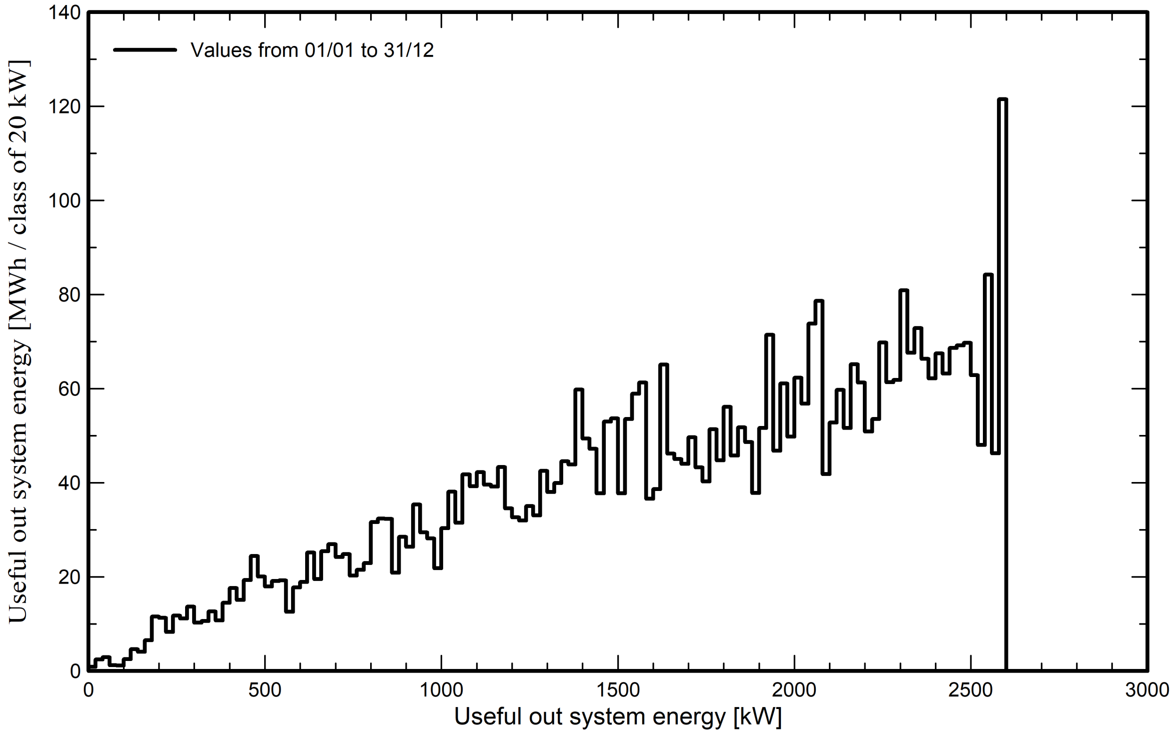


Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



P50 - P90 evaluation

Weather data

Source	Meteonorm 8.2 (2006-2013), Sat=100%
Kind	Monthly averages
Synthetic - Multi-year average	
Year-to-year variability(Variance)	7.1 %

Specified Deviation

Climate change	0.0 %

Global variability (weather data + system)

Variability (Quadratic sum)	7.3 %

Simulation and parameters uncertainties

PV module modelling/parameters	1.0 %
Inverter efficiency uncertainty	0.5 %
Soiling and mismatch uncertainties	1.0 %
Degradation uncertainty	1.0 %

Annual production probability

Variability	371 MWh
P50	5050 MWh
P90	4574 MWh
P95	4440 MWh

Probability distribution